

Vedprakash Meena

Bharatpur, Rajasthan, India

+91 7023083405 • vedumeena01@gmail.com • linkedin.com/in/vedprakash-meena-58a427309 • github.com/vedumeena01

Professional Summary

Software Engineer and AI Model Trainer with strong expertise in **Python, C++, Data Structures & Algorithms, and LLM Evaluation/RLHF**. Experienced in analyzing AI model responses, authoring adversarial coding prompts, step-by-step mathematical reasoning, and providing gold-standard reference implementations. Certified by NASSCOM in AI Data Engineering workflows with a solid background in Computer Science fundamentals (OOPs, DBMS, OS, Networks). Passionate about advancing frontier AI systems through high-quality training data, multi-turn reasoning rubrics, and rigorous code verification.

Technical Skills

Programming Languages:	Python 3.x, C, C++, Java , JavaScript (ES6+), SQL, Bash / Shell Scripting
AI & Model Alignment:	LLM Evaluation, RLHF (Reinforcement Learning from Human Feedback) , Adversarial Prompting, Model Benchmarking, Multi-Turn Reasoning, Step-by-Step Chain-of-Thought (CoT)
Core Computer Science:	Data Structures & Algorithms (200+ Solved) , Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks
Frameworks & Tools:	FastAPI, Flask, React.js, Node.js, PyTorch, OpenCV, Git, GitHub, Docker, VS Code, Postman, Linux
Languages & Writing:	Fluent in English and Hindi (Technical Writing, Code Explanations, Linguistic Nuance, Bilingual Translation)

Technical Projects & Experience

1. AI Video-to-Text Multimodal Captioning Pipeline Jan 2026 – March 2026

Tech Stack: PyTorch, Transformer Decoder, ResNet-152, C3D, Python, CUDA

- Engineered an end-to-end multimodal deep learning pipeline extracting spatiotemporal features from video keyframes to generate coherent, grammatically precise natural language descriptions.
- Built a dual-backbone feature extractor (ResNet-152 for spatial context + C3D for temporal dynamics) with a cross-modal projection layer.
- Integrated an autoregressive Transformer decoder with hybrid Cross-Entropy and Contrastive Loss, boosting semantic alignment and BLEU-4 metrics.

2. Real-Time AI Biometric & Anti-Spoofing Verification System Dec 2025 – Jan 2026

Tech Stack: Python, OpenCV, PyTorch, dlib, Silent-Face Anti-Spoofing, AES-256 Encryption

- Developed a real-time computer vision security pipeline integrating multi-face detection with a deep neural network anti-spoofing ensemble to reject replay attacks and photo spoofs.
- Engineered a thread-safe encrypted database utilizing AES-256 Fernet symmetric encryption for secure biometric storage and sub-50ms identification latency.
- Optimized multi-threaded frame processing, achieving 30+ FPS real-time throughput on live video feeds with 97.8% anti-spoofing accuracy.

3. Autonomous Line-Following & Dynamic Maze-Solving System Jan 2026 – Feb 2026

Tech Stack: C, C++, Microcontrollers, Sensor Fusion, Closed-Loop PID Control, Graph/Maze Algorithms

- Architected an autonomous robotic navigation system utilizing sensor arrays and closed-loop **PID controllers** for high-speed line tracking.
- Engineered dynamic maze-solving algorithms (Left-Hand Rule & Shortest Path graph traversal) to detect dead ends and compute optimal return paths in real-time.
- Implemented low-level motor PWM calibrations and hardware interrupts, reducing response latency by 35%.

4. DevFlow — Full-Stack Collaborative Project & Task Platform 2025

Tech Stack: React.js, Node.js, Express.js, MongoDB Atlas, JWT Authentication, REST APIs

- Built a responsive collaborative project management web application featuring JWT authentication and role-based access control.
- Designed 15+ RESTful API endpoints on Express.js with robust payload validation, centralized error handling, and MongoDB Atlas schemas.

Education

Indian Institute of Information Technology (IIIT), Sonapat Aug 2023 – Aug 2027

Bachelor of Technology (B.Tech) in Information Technology — Sonapat, Haryana

Adarsh Regional Senior Secondary School 2022

RBSE Class XII — **83.0%** — Nadbai, Bharatpur, Rajasthan

Certifications & Achievements

- AI - Data Engineering Analyst:** Certified by **NASSCOM / IT-ITes SSC & Skill India Digital Hub** in AI data engineering concepts and workflows (Aug 2026).
- Mastering Power BI:** Certified by **PHN Technology & Skill India Digital Hub** in data modeling and analytics (Aug 2026).
- Algorithmic Problem Solving:** Solved **200+ Data Structures & Algorithms problems** across LeetCode and GeeksforGeeks (Arrays, Trees, Graphs, DP).
- Bilingual AI Trainer:** Strong capability in technical reasoning, prompt engineering, and comparative evaluations in English and Hindi.